

Energy Market Trade Optimization & Shadow Settlement Platform

Student-led | Team Lead: Ruhi Kabarwal | Partner: Southern Company | Manager: Veronica Powell | 2026-2027

This is a **student-led** Senior Design project led by **Ruhi Kabarwal**. We are forming a four-person team to build a proof of concept (POC) in collaboration with **Southern Company**, one of the largest electric utilities in the United States. We are collaborating with **Veronica Powell**, who manages the team at Southern Company, and we will meet **weekly with company liaisons** who will guide requirements, review our work, and help us align the POC with Southern Company's real infrastructure. We are looking for teammates who want hands-on experience with enterprise data platforms, who have an interest in **emerging technologies** (AI assistants, multi-agent systems, modern backends), and who would like to gain experience working with business needs and pitching new technologies to the company.

1. Background

Wholesale electric utilities operate in volatile deregulated energy markets such as PJM and ERCOT. Every hour, commercial operators decide whether to sell generation into the **Day-Ahead Market** (predictable, lower risk) or the **Real-Time Market** (volatile, potentially higher reward). At the same time, Regional Transmission Organizations (RTOs) send massive, complex daily billing statements. Most utilities have no automated way to independently verify these multi-million dollar invoices against their own asset telemetry and trade positions. Undetected billing errors become **revenue leakage**.

2. Project Needs

- **Shadow Settlement:** Independently recalculate what the utility should have been billed and flag discrepancies above a dollar threshold.
- **Trade Optimization:** Analyze Day-Ahead vs. Real-Time prices to show where trading decisions could have captured more revenue.
- **Explainability:** Turn raw variance numbers into plain-language explanations and ready-to-send dispute documents.
- **Enterprise Fit:** Build on tooling that maps to Southern Company's current infrastructure so the POC has a credible path to adoption.

3. Project Scope

The team will deliver a working, end-to-end platform on simulated data consisting of:

- **Grid Simulator:** A Python generator streaming realistic Locational Marginal Prices (LMP), plant output (MW), and internal trade logs, since real utility data is proprietary.
- **Data Lakehouse:** Databricks on AWS or Azure using the Medallion pattern (Bronze, Silver, Gold) with Auto Loader, PySpark, Delta Live Tables, and Unity Catalog.
- **Shadow Settlement Engine:** PySpark/SQL modules that compute expected costs and isolate variances against simulated RTO bills.
- **Arbitrage Module:** An analytics or lightweight ML forecasting model (MLflow) that outputs a daily optimization strategy for operators.
- **AI Settlement Assistant:** A Claude API powered assistant that explains each flagged variance in plain language and drafts dispute documents, with a multi-agent design that coordinates research, explanation, and drafting agents.
- **Operator Dashboard:** A web interface for visualizing revenue leakage, reviewing market positions, drilling into discrepancies, and filing disputes.

4. Technologies: Core Stack and Emerging Tools

Part of our goal is to work with emerging technologies. Alongside the core stack, we intend to integrate the following newer tools into the workflow and evaluate how they fit Southern Company's environment:

Core platform	Databricks (Auto Loader, PySpark, Delta Live Tables, Unity Catalog), serverless SQL Warehouses, Python backend, GitHub Actions CI/CD, Databricks Apps / Power BI or React for the front end
AI assistant	Claude API for variance explanation and dispute-document drafting
Agent memory	Honcho for persistent, user-aware memory across agent sessions
Multi-agent setup	Postgres for state and Redis for task queues and message passing between agents
Application backend	Supabase (Postgres, auth, storage, realtime) serving the web application
Design	Figma for wireframes, dashboard mockups, and operator workflow design

Technical challenges include processing minute- and hour-level time series across dozens of simulated assets (partitioning, shuffling, cluster tuning), keeping settlement math auditable, designing reliable agent handoffs, and grounding the AI assistant in Gold-layer data so its explanations are accurate.